

TIN DANG HUYNH TRUNG

Senior Software Engineer – LLM Infrastructure, Backend & Distributed Systems

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Eight years building production backend systems in Python, Rust and TypeScript, with machine learning in production since 2019 and the last two years on the infrastructure under AI products: multi-provider LLM gateways, routing with failover, cost metering that reconciles against what providers actually bill, agent memory and MCP. I came to this from fintech and build it the same way – correctness enforced by structure rather than convention: append-only ledgers, multi-tenancy that fails closed, money that cannot move twice.

PROFESSIONAL EXPERIENCE

Senior Software Engineer / Technical Lead

Trustify Technology

Jan 2024 – Present

Ho Chi Minh City, Vietnam

I architect and lead client rebuilds across Vietnam and Myanmar – fintech, real estate and intelligent transport for Yoma Group, Yoma Bank, Wave Money and Trustonic – and wrote the company's engineering career ladder. Alongside the work below I built Yoma Bank's microfinance lending MVP in Java 21 and Spring Boot, and owned the money and authentication paths on Star City's resident platform for a team of nine.

- **Estate-finance platform** • *Yoma Group – sole engineer & architect*

The legacy system kept balances in rows anyone could edit, so I rebuilt it off PHP/Laravel onto a single append-only double-entry ledger where every balance can be re-derived and audited, with per-customer cutover, an outbox and reconciliation around it. Before anyone moved I graded 235 legacy business rules against the new system and published the gaps.

Python 3.12 • FastAPI • async SQLAlchemy 2 • PostgreSQL (schema-per-tenant) • Redis • Next.js • AWS ECS • Envoy

- **Card payments & PCI boundary** • *Yoma Group, Wave Money*

I put cardholder data behind its own service, database and network so that only that service falls inside PCI scope, and made the build reject any import that crosses the boundary. Card payments run through 2C2P with callbacks that are safe to retry and a three-way reconciliation behind them.

Python • FastAPI • PostgreSQL • 2C2P • Docker

- **Real-time ML serving at the edge** • *Ho Chi Minh City – architect & sole engineer*

400 camera streams produce more video than a city network can carry, so I moved inference to the edge and sent only events onward: YOLO models served on TensorRT and Triton hold a P95 of 85 ms, Jetson nodes cut backhaul by 75%, and events reach the data centre over mutually authenticated Kafka. I load-tested the pipeline to 71K events a second before adopting it.

Python • FastAPI • Kafka • PyFlink • TensorRT • Triton • K3s • ArgoCD • Jetson • DuckDB

- **Device-management migration** • *Trustonic*

A customer could only be moved if they could be moved back, so I owned the rollback across PostgreSQL and DocumentDB. Rehearsing it is what found the gap: Postgres reverted cleanly, the document store, cache and archived objects did not, so a real rollback would have left the customer internally inconsistent. I extended the rollback to cover them, and tuned the migration jobs: vectorised batch validation, broadcast joins in place of shuffles, keyed DynamoDB reads so a run never scans a live table, and Redis partitioning chosen by measurement.

PySpark • AWS Glue • Airflow • DynamoDB • PostgreSQL • DocumentDB • Redis • boto3

CTO – Co-founder

ClickAI Viet Nam

Jun 2024 – Oct 2025 (concurrent)

Ho Chi Minh City, Vietnam

I co-founded ClickAI (clickai.vn) and built its conversational AI platform from scratch – multi-modal chat, image and audio generation, community – to 40K+ users across Southeast Asia inside four months. As CTO I set the architecture and engineering standards, led a team of twelve to fourteen, and owned billing, identity and deployment myself.

- **Model platform** • *20+ providers*

I built the orchestration layer that routes across 20 third-party providers and one of our own, with failover, vector memory and reasoning-model support; choosing the model by policy rather than habit cut API spend by 20%. The first time a provider went hard-down, users hit the failure before routing shifted – so I stopped waiting for failures and probed providers continuously, marking one down before traffic could reach it. Generation fans out to ComfyUI, Replicate, fal.ai and ElevenLabs behind circuit breakers.

Python • FastAPI • Weaviate • Qdrant • FalkorDB

- **Billing & metering** • *sole author*

I built the wallet, plan, promotion and referral-commission engine over a month-partitioned ledger that meters every LLM call. A retry loop was quietly re-sending the same work, and a reconciliation check I wrote is what caught it – provider billing had drifted more than 30% from what we metered. I corrected the ledger model and its indexes, keyed idempotency on the source event so a retry cannot double-count, and made the check a standing job.

Python • FastAPI • PostgreSQL (partitioned) • Redis

- **Production, identity & web** • 99.9% uptime, 1K+ concurrent sessions

I authored the 13-service deployment with cert-manager and autoscaling, and held sub-2s responses. When the database saturated I moved PostgreSQL onto a three-node Patroni cluster; when a release regressed, my own instrumentation caught it before any customer did; and a hosting comparison I took to stakeholders halved the monthly bill.

Kubernetes · Helm · Azure · Patroni · Prometheus · Next.js 14 · TypeScript

Founder & Principal Engineer

Pilot Space (open-source AI infrastructure)

📅 Jan 2026 – Present

📍 Remote

- **Hydroa** • multi-tenant LLM gateway

A gateway that meters what each tenant spends on LLMs. Envoy authorises at the edge, six providers sit behind per-tenant and per-provider circuit breakers, customer keys are encrypted at rest, and a cost ledger reconciles against what the providers actually billed.

Python · FastAPI · Envoy · PostgreSQL · Kubernetes · OpenTelemetry

- **Lunaris & Helios** • agent memory and agent harness

Lunaris recalls across 100K documents per scope in 19–22 ms at the median, and a CI gate fails the build on any 5% regression. Helios is the harness that consumes it: 12 provider backends, a native MCP client and server, WASM-sandboxed extensions, and a circuit breaker on every outbound call.

Rust · Tokio · MCP · Wasmtime · npm

- **Moon** • Redis-compatible in-memory store, written from scratch

I wrote it from scratch so that existing Redis clients connect unchanged: RESP2/RESP3, 250+ commands, and native vector, BM25 and graph search under MVCC. It has shipped 30 releases to v0.8.9 and measures 1.62× on GET and 1.32× on SET against Redis 7.0 with 8–22% less memory per key, published alongside the cases where it loses.

Rust · Tokio · io_uring · crates.io · PyPI

- **ADD** • published development method

Specify the work, write the failing tests, build until they pass, then verify against a recorded result rather than a plausible diff. I published a six-milestone benchmark of it, then retracted my own headline claim and republished after finding a defect in my own meter. Shipped as @pilotspace/add and pilotspace-add at 3.6.0, about 1.2K downloads a month.

EARLIER EXPERIENCE (2018 – 2024)

- **Senior Data Engineer**, Asia Commercial Bank (ACB) • Aug 2023 – Jun 2024

Ran an enterprise data lake carrying 5M+ daily transactions. A data-quality framework I built across six dimensions cut manual verification by 80%, and my APIs served 100+ internal applications. Cloudera · PySpark · Dagster · Iceberg · FastAPI

- **Software Data Engineer**, Vietmoney Corp • Mar 2021 – Aug 2023

Built on a fintech appraisal and lending platform running 10M+ daily transactions at 99.9% uptime. Price forecasting at 96% accuracy cut appraisal from 15 minutes to 5 seconds, and a distributed crawler doubled throughput across 50+ SOURCES. Python · PyTorch · BERT · Scrapy · Pub/Sub · BigQuery · Airflow · GCP

- **Software Developer**, Dai Phat Solutions • Aug 2019 – Mar 2021

Shipped CCTV analytics at 15 FPS to 20+ clients, cutting reported incidents by 40%, and face recognition at 92% accuracy over 1,000+ employees that removed 90% of a manual process. Python · PyTorch · OpenCV · FAISS · Kafka · FFmpeg

- **Full-stack Developer**, LogiGear Corporation • Jun 2018 – Mar 2019

Built Ethereum wallet, smart-contract and liquidity-pool management in Node.js and React.

OPEN SOURCE

- **yowo** — production YOLO11/YOLO26 inference and export across TensorRT, ONNX Runtime, CoreML and OpenVINO, with ByteTrack and CLIP-ReID tracking and hardware-aware backend selection; on PyPI. **Moon**, **Lunaris** and **ADD** ship from github.com/pilotspace through crates.io, npm and PyPI. Also **fastapi-multi-tenant**; **rjson**, Rust-backed JSON for Python; **face-anti-spoofing**, founded at Vietmoney.

TECHNICAL SKILLS

AI & LLM infrastructure: Multi-provider gateways, policy routing & failover, token metering & cost reconciliation, agent memory, MCP (client & server), RAG, vector search (Weaviate, Qdrant, FalkorDB); PyTorch, YOLO, TensorRT, Triton, ONNX Runtime, OpenVINO

Languages & backend: Python (typed, async), Rust, TypeScript / Node.js, Java, SQL; FastAPI, SQLAlchemy 2, Tokio, Next.js / React; clean & hexagonal architecture, CQRS, event sourcing, multi-tenancy, outbox & idempotency, PCI boundaries, OAuth / OIDC

Data, storage & cloud: PostgreSQL (Patroni HA, RLS, partitioning), Redis, MongoDB, DynamoDB, DuckDB; Kafka, PyFlink, PySpark, AWS Glue, Airflow, BigQuery; Kubernetes, Helm, Envoy, OpenTelemetry, AWS, Azure, GCP

EDUCATION, TEACHING & CERTIFICATIONS

BSc Computer Science (High-Quality Program), Ton Duc Thang University, 2015–2019. **Teaching:** Design Patterns instructor, Ton Duc Thang University (2022); mentor, MindX (2023); speaker, AI Workshop, FPT University (2025). **Certifications:** Google Cloud AI Agents (2025); AWS ML Foundations (2022). **Languages:** English, Vietnamese.